Name: Viraj. M. Bhosale Class: SOC-09 Batch: C
Subject: PL-II (Python) Roll No.: 61 Exp. No.: 02
Name of the Experiment: Sequence - Types of and operator.
Performed on: 10/02/2026.
Submitted on: 16/02/2026.

Marks	Teacher's Signature with date
10	

1) List the python sequence data types and state one key characteristics of each with respect to mutability?

→

i) List = ordered and mutable (elements can be changed)
ii) Tuple = ordered and immutable (cannot be modified)
3) String = ordered and immutable sequence of characters.
iv) set = unordered, mutable, store unique elements only
v) Dictionary.

2) Explain how indexing and slicing work in python sequences Illustrate how they differ when applied to string and list?

→ Indexing = It is used to access a single element using its position (start from 0).

Incomplete for Theory ☐ Diagrams ☐ Observation Table ☐
Calculations ☐ Graphs ☐ Results ☐ Conclusion ☐
Understanding Through Q/A ☐ Late Submission ☐ Neatness ☐

Teacher's Signature

Slicing: It is used to acess a part of the sequence using [Start : end : step].

Ex: S="Python"
I= [10, 20, 30, 40].

* Difference
→ String cannot be modified after slicing (immutable).
→ List can be modified after slicing (mutable).

Q3) Write a python program that.
→ a) uses of indexing and slicing on a string list and
b) applies atleast two built in function to apply display meaning full result.

```
→ text= "Viraj".
print ("first Character:", text[0]).
print ("Substring :", text [0:3]).
print(" length of string:", len(text)".
print ("upper case:", text upper()).
```

Output :-
First Character : V
Substring : VidVir
Length of String : 7.
upper case : Viraj

4)

→ Equality (==) Checks wheather values are same.
Identity (is) checks wheather both variable point to the same memory location.
Ex,

```
a = [1,2,3]
b = [1,2,3]
print = [a == b]
print = [a is b]
output :-
True
false
```

Explations →
- Lists are mutable and stored in diffrent memory location.
- (==) compares content.
- (is) componentns memory reference.

5)

→ List = Allows duplicates, slower searching
Set = Stores only unique, value no indexing
Dictonaries :- Stores unique student. IDs as keys and detail as value.
Most efficient = Dictonary, because it allow fast access, unique keys and structure data storage.

6)

→ Students = {101, Amit, 102, "Neha", 103, "Rahul"}.
print ("Students 101 :", students [101).
print ("Total Students : len (Students)).
print (" Students ID'S :") List (Students key (1)).

Output.

Student 101: Amit.

Total student : 3

Student ID'S = [101, 102, 103].

7)

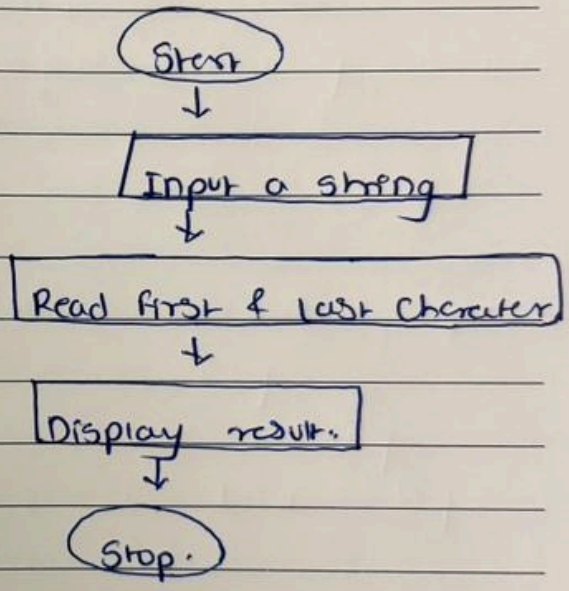
→ S = input ("Enter a String:").
print ("First Characteristic :" S[0]).
print ("Last Character:" S[-1])

→ output

: First Character : P.
Last Character : n.

Algorithm

1) Start.
2) Read a string
3) Find the First Character
4) Find the last Character
5) Display result.
6 Stop



8)

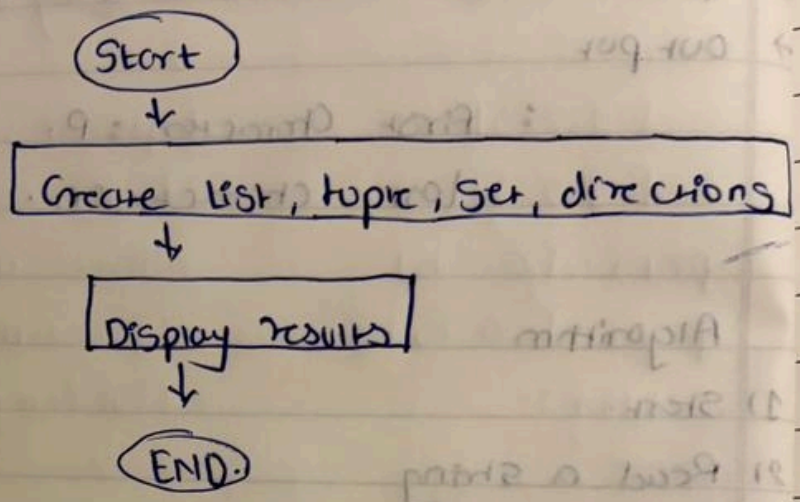
→ List = [1, 2, 3]
tuple = [4, 5, 6]
String = [7, 8, 9]
dictonary = [1: One", 2 : "Two")
print (list, tuple, String, dictonary).

→ Algorithm :- 1) Start
2) Create a list, tuple, set, dictonary.
3) Display result.
4) Stop.

Output

→ [1, 2, 3]

[4, 5, 6].



9)

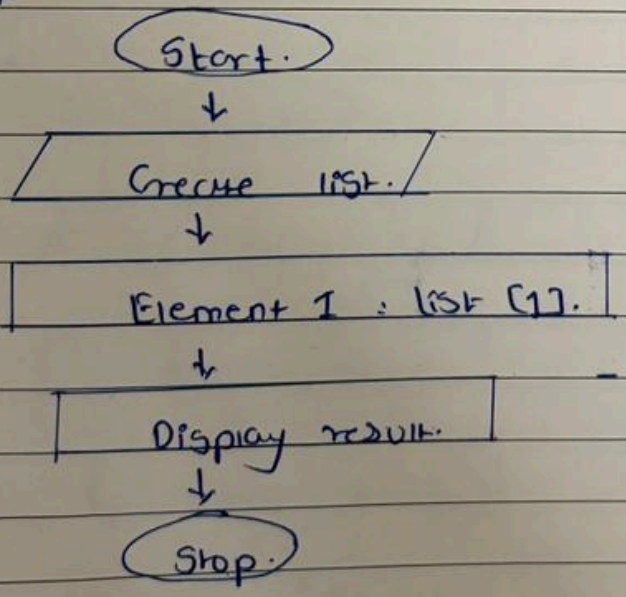
→ Array = [10, 20, 30, 40]
print (" Element at index 1: array [1]]
print ("Element at index -1 "array [-1]]

→

Output → Element at index 1: 20
Element at index 1: 40.

* Algorithm
i) Start
ii) Create a list of element.
3) Access element
4) Display the result
5) Stop.

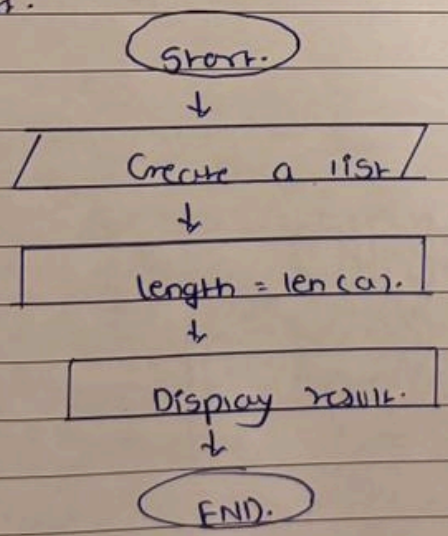
* Flowchart



10)

→ num = (5, 2, 9, 1).
print ("length:", len (num)).
print (maximum :" max (num).
print ("sorted list:" sorted num)

→ output = length : 4
Maximum : 9
Minimum : 1.
Sorted list :- [1, 2, 5, 9]

Algo :- 1) Start
2) Creates list
3) Use len() function, max () and minimum().
4) END.

Flow Chart.